King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 102 Final Exam Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. The series $\sum_{n=0}^{\infty} \frac{(x-2)^n}{n^2+1}$ converges for

(a) $x \in [1, 3]$

(b) $x \in [1, 4]$

(c) $x \in (1, 3]$ only

(d) $x \in [1, 3)$ only

(e) all values of x

2. The area of the region bounded by $y = x^2 \ln x$ and $y = 4 \ln x$ is equal to:

(a) $\frac{16}{3} \ln 2 - \frac{29}{9}$

(b) $\frac{8}{3}e - \frac{1}{2}e$

(c) $\frac{e^2}{2} - e \ln 2$

(d) $\frac{16 \ln 2 - 3}{9}$

(e) $2 \ln 2 - 1$

3. The series $\sum_{n=1}^{\infty} \frac{10^n}{(n+1)3^{2n+1}}$

(a) converge by integral test

(b) converge conditionally

(c) converge by the root test

(d) converges by the ratio test

(e) diverges by the ratio test

4. The length of $y = \ln(1 - x^2)$ over $\left[0, \frac{1}{2}\right]$ is

(a) $\ln 3 - \frac{1}{2}$

(b) $\frac{1}{2}\ln(1 + e)$

(c) $2 \ln 3$

(d) $\ln(e - 1)$

(e) $\ln 3 - \ln 2$

5. $\int \frac{1}{\sqrt{x} - \sqrt[3]{x}}\, dx =$ (Hint: Substitute $u = \sqrt[6]{x}$)

(a) $\sqrt{x} - \frac{3}{2}\sqrt[4]{x} + 3 \ln\left|\sqrt[6]{x - 1}\right| + c$

(b) $\sqrt[4]{x} - \frac{3}{2}\sqrt[3]{x} + 6 \ln\left|\sqrt[3]{x} - \frac{1}{2}\right| + c$

(c) $2\sqrt{x} + 3\sqrt[3]{x} + 6\sqrt[6]{x} + 6 \ln|\sqrt[6]{x} - 1| + c$

(d) $6\sqrt[3]{x} + 6 \ln|1 - \sqrt[4]{x}| + c$

(e) $\sqrt{x}\sqrt[6]{x - 1} + 3\sqrt[4]{x} + \frac{1}{6}\ln\left|\sqrt[3]{x - 1}\right| + c$

6. The series $\sum_{n=1}^{\infty} \frac{\operatorname{arc\,cot} n}{n^{2.1}}$

(a) converges by comparing with $\pi \sum_{n=1}^{\infty} \frac{1}{n^{2.1}}$

(b) diverges by comparing with $\pi \sum_{n=1}^{\infty} \frac{1}{n^{0.1}}$

(c) diverges by limit comparison test

(d) converges by comparing with $\pi \sum_{n=1}^{\infty} \frac{1}{n^{0.1}}$

(e) diverges by comparing with $\pi \sum_{n=1}^{\infty} \frac{1}{n^{2.1}}$

7. The surface area that obtained when the curve $y = \sin(\pi x)$, $0 \le x \le 1$ is revolved about the x-axis is equal to:

(a) $\frac{2}{\pi}\ln\left(\pi^2 + \sqrt{1 + \pi}\right)$

(b) $\frac{1}{\pi}\ln\left(\pi + \sqrt{\pi^2 + \pi}\right)$

(c) $\frac{2}{\pi}\ln\left(\sqrt{\pi + \sqrt{1 + \pi^2}}\right)$

(d) $2\sqrt{1 + \pi^2} + \frac{2}{\pi}\ln\left(\pi + \sqrt{1 + \pi^2}\right)$

(e) $\frac{4}{\pi}\sqrt{1 + \pi^2} + \ln\left(\sqrt{1 + \pi^2}\right)$

8. The series $\sum_{n=1}^{\infty} \frac{n \cos n\pi}{3^n}$ is

(a) a divergent alternating series

(b) a convergent alternating series

(c) a convergent geometric series

(d) a divergent p-series

(e) a divergent geometric series

9. $\int_0^1 \frac{dx}{(1+\sqrt{x})^4} =$

(a) $\frac{1}{16}$

(b) $\frac{2}{6}$

(c) $\frac{3}{8}$

(d) $\frac{3}{16}$

(e) $\frac{1}{6}$

10. $\int_0^{\infty} x^2 e^{-x^2}\, dx =$

(a) $\frac{1}{2}\int_0^{\infty} xe^{-x}\, dx$

(b) $\frac{1}{2}\int_0^{\infty} e^{-x^2}\, dx$

(c) 0

(d) $2\int_0^{\infty} xe^{-x^2}\, dx$

(e) $\int_{-\infty}^{\infty} xe^{-x^2}\, dx$

11. Given that the base of a solid is the region bounded by the parabolas $y = x^2$ and $y = 2 - x^2$. The volume of the solid if the cross-sections perpendicular to the x-axis are squares with one side lying along the base is:

(a) $\frac{28}{15}$

(b) $\frac{64}{15}$

(c) $\frac{8}{15}$

(d) $\frac{32}{15}$

(e) $\frac{16}{15}$

12. The sequence $a_n = \dfrac{\sin 2n}{1+\sqrt{n}}$

(a) converges to 1

(b) converges to 0

(c) diverges

(d) converges to -1

(e) converges to 2

13. $\int \frac{dx}{\sqrt{e^x-1}} =$

(a) $\dfrac{e}{2}\tan^{-1}\left(\sqrt{e^x-1}\right)+c$

(b) $\dfrac{2}{e}\tan^{-1}\left(\dfrac{x}{2}\right)+c$

(c) $\dfrac{1}{2}\tan^{-1}\left(\dfrac{1}{2}x\right)+c$

(d) $\dfrac{2}{e}\tan^{-1}\left(\sqrt{e^x-1}\right)+c$

(e) $2\tan^{-1}\left(\sqrt{e^x-1}\right)+c$

14. Using the method of cylindrical shell, the volume generated by rotating the region bounded by $y = x^3$, $y = 0$, $x = 1$ about $y = 1$ is equal to:

(a) $\dfrac{5\pi}{14}$

(b) $\dfrac{10\pi}{14}$

(c) $\dfrac{16}{14}\pi$

(d) $\dfrac{13}{14}\pi$

(e) $\dfrac{3}{14}\pi$

15. The series $\sum_{i=1}^{\infty} \frac{\ln n}{n^2}$

(a) diverges by integral test $\int_1^{\infty} \frac{\ln x}{x^2}\,dx = \infty$

(b) converges by integral test $\int_1^{\infty} \frac{\ln x}{x^2}\,dx = \dfrac{1}{2}$

(c) diverges by integral test $\int_1^{\infty} \frac{\ln x}{x^2}\,dx = -\infty$

(d) converges by integral test $\int_1^{\infty} \frac{\ln x}{x^2}\,dx = 1$

(e) converges by integral test $\int_1^{\infty} \frac{\ln x}{x^2}\,dx = \dfrac{1}{8}$

16. The values of p for which the series $\sum_{n=1}^{\infty} n\left(\left(1+(2n)^2\right)^{\frac{p}{2}}\right)$ is convergent.

(a) $p<1$

(b) $p<0$

(c) $p>-2$

(d) $p<-2$

(e) $p<-1$

17. The Maclaurin Series for $f(x)=\sinh x$ is

(a) $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$

(b) $\sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n)!}$

(c) $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!}$

(d) $\sum_{n=0}^{\infty}\frac{x^{2n}}{(2n)!}$

(e) $\sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n+1)!}$

18. The series $\sum_{n=1}^{\infty}a_n$ where $a_n=\dfrac{3n^3+n}{\sqrt{4+n^7}}$

(a) converges by limit comparison test with $\left(b_n=\dfrac{3}{n^{\frac{3}{2}}}\right)$

(b) diverges by limit comparison test with $\left(b_n=\dfrac{3}{n^{\frac{1}{2}}}\right)$

(c) converges by limit comparison test with $\left(b_n=\dfrac{3}{n^{\frac{1}{2}}}\right)$

(d) diverges by limit comparison test with $\left(b_n=\dfrac{3}{n^{\frac{7}{2}}}\right)$

(e) converges by limit comparison test with $\left(b_n=\dfrac{3}{n}\right)$

19. The series $\sum_{n=1}^{\infty}\ln\left(1+\frac{1}{n}\right)$ is

(a) converges to $\dfrac{1}{2}$

(b) diverges

(c) converges to 3

(d) converges to 0

(e) converges to 2

20. $\int_0^3 |x^2 - 4|\, dx =$

(a) $\dfrac{29}{3}$

(b) $\dfrac{23}{3}$

(c) 21

(d) $\dfrac{16}{3}$

(e) $\dfrac{17}{2}$

21. A formula for the general term a_n of the sequence $\left\{\dfrac{3}{5}, \dfrac{-4}{25}, \dfrac{5}{125}, -\dfrac{6}{625}, \dfrac{7}{3125}, \ldots\right\}$ is

(a) $(-1)^n \dfrac{n+2}{5^n}$, $n > 1$

(b) $(-1)^n \dfrac{n+3}{5^n}$, $n \geq 0$

(c) $(-1)^{n-1} \dfrac{n+3}{5^{n+1}}$, $n \geq 0$

(d) $(-1)^{n-1} \dfrac{n+2}{5^n}$, $n \geq 1$

(e) $(-1)^{n-1} \dfrac{n+2}{5^n}$, $n \geq 0$

22. The average value of $f(x) = \sin^2 x \cos^3 x$ over $[-\pi, \pi]$ is equal to:

(a) $\dfrac{1}{15}\left(\pi^3 - \pi^5\right)$

(b) $\dfrac{1}{15\pi}$

(c) $\dfrac{1}{2\pi}$

(d) $\dfrac{\pi}{15}\left(\pi^5 - \pi^3\right)$

(e) 0

23. The series $\sum_{n=1}^{\infty}\left(\frac{2}{e^n}+\frac{1}{2n(n+1)}\right)$

(a) converges and the sum is $\dfrac{3+e}{2e-2}$

(b) converges and the sum is $\dfrac{2}{e-1}$

(c) converges and the sum is $\dfrac{e}{e-1}$

(d) converges and the sum is $\dfrac{1}{e-1}$

(e) converges and the sum is $\dfrac{e+1}{e-1}$

24. $\int_1^9 \frac{2x^2+x^2\sqrt{x}-1}{x^2}\,dx$

(a) $29\dfrac{1}{4}$

(b) $32\dfrac{4}{9}$

(c) $33\dfrac{3}{4}$

(d) $34\dfrac{1}{2}$

(e) $36\dfrac{1}{4}$

25. The power series representation for the function $\dfrac{x}{9+x^2}$ is

(a) $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{9^n}$

(b) $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{3^{n+1}}$

(c) $\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{9^{n+1}}$

(d) $\sum_{n=0}^{\infty}\frac{x^{2n+1}}{9^{n+1}}$

(e) $\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{9^n}$

26. $\sin^2 x =$

(a) $1 - \left(\sum_{n=0}^{\infty} \dfrac{(-1)^n x^{2n+1}}{(2n+1)!}\right)^2$

(b) $\dfrac{2}{2!}x^2 - \dfrac{2^3}{4!}x^4 + \dfrac{2^5}{6!}x^6 - \dfrac{2^7}{8!}x^8 + \cdots$

(c) $1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dfrac{x^6}{6!} + \dfrac{x^8}{8!} - \cdots$

(d) $x^2 + \dfrac{x^6}{(3!)^2} + \dfrac{x^{10}}{(5!)^2} + \dfrac{x^4}{(7!)^2} + \cdots$

(e) $\dfrac{1}{2} - \dfrac{2}{2!}x^2 + \dfrac{2^3}{4!}x^4 - \dfrac{2^5}{6!}x^6 + \cdots$

27. $\lim_{h\to 0} \frac{1}{h} \int_2^{2+h} \sqrt{1+t^3}\, dt =$

(a) $\sqrt{3}$

(b) $\sqrt{2}$

(c) 4

(d) 3

(e) 0

28. The series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{n!(n+1)!\, 2^{2n+1}}$ converges for

(a) $[1, 3]$ only

(b) all values of x

(c) $(-\infty, 0)$ only

(d) $(0, \infty)$ only

(e) $(1, 3]$ only

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(a)**	$R = 1$ centered at 2; both endpoints converge (compare with $\sum 1/n^2$): $[1, 3]$.
2	**(a)**	Limits $x = 1, 2$; $\int_1^2 (4 - x^2) \ln x \, dx = \frac{16}{3} \ln 2 - \frac{29}{9}$.
3	**(e)**	Ratio $\to \frac{10}{9} > 1$, so it diverges by the ratio test.
4	**(a)**	$\sqrt{1 + y'^2} = \frac{1+x^2}{1-x^2} = -1 + \frac{2}{1-x^2}$, so $L = \ln 3 - \frac{1}{2}$.
5	**(c)**	$u = \sqrt[6]{x}$: $\int \frac{6u^3}{u-1} du = 2u^3 + 3u^2 + 6u + 6 \ln \lvert u - 1 \rvert + c$.
6	**(a)**	$0 < \operatorname{arc\,cot} n < \pi$, compare with convergent p-series ($p = 2.1$): converges.
7	**(d)**	$u = \pi \cos \pi x$: $S = \frac{4}{\pi} \int_0^{\pi} \sqrt{1 + u^2} du = 2\sqrt{1 + \pi^2} + \frac{2}{\pi} \ln(\pi + \sqrt{1 + \pi^2})$.
8	**(b)**	$\cos n\pi = (-1)^n$; $b_n = n/3^n$ decreases to 0: convergent alternating series.
9	**(e)**	$u = \sqrt{x}$, $w = 1 + u$: $2 \int_1^2 (w^{-3} - w^{-4}) dw = \frac{1}{6}$.
10	**(b)**	By parts ($u = x$, $dv = xe^{-x^2} dx$): the boundary term vanishes, leaving $\frac{1}{2} \int_0^{\infty} e^{-x^2} dx$.
11	**(b)**	Side $2 - 2x^2$ on $[-1, 1]$: $V = 4 \int_{-1}^{1} (1 - x^2)^2 dx = \frac{64}{15}$.
12	**(b)**	$\lvert a_n \rvert \le \frac{1}{1+\sqrt{n}} \to 0$, so $a_n \to 0$ by squeeze.
13	**(e)**	$u = \sqrt{e^x - 1}$: $dx = \frac{2u \, du}{1+u^2}$, so the integral is $2 \tan^{-1} u + c$.
14	**(a)**	Radius $1 - y$, height $1 - y^{1/3}$: $V = 2\pi \int_0^1 (1 - y)(1 - y^{1/3}) dy = \frac{5\pi}{14}$.

Q	Ans	Brief explanation
15	**(d)**	By parts $\int_1^\infty \frac{\ln x}{x^2}dx = 1$ (finite), so the series converges.
16	**(d)**	Term $\sim 2^p n^{p+1}$; a p-series with exponent $-p-1>1$ converges, so $p<-2$.
17	**(e)**	$\sinh x = \frac{e^x - e^{-x}}{2}$ keeps odd powers: $\sum \frac{x^{2n+1}}{(2n+1)!}$.
18	**(b)**	$a_n \sim \frac{3}{n^{1/2}}$, a divergent p-series ($p=\frac{1}{2}$), so it diverges.
19	**(b)**	Telescopes to $S_N = \ln(N+1) \to \infty$, so it diverges.
20	**(b)**	Split at $x=2$: $\frac{16}{3} + \frac{7}{3} = \frac{23}{3}$.
21	**(d)**	$a_n = (-1)^{n-1}\frac{n+2}{5^n}$ starting at $n=1$ ($a_1 = \frac{3}{5}$).
22	**(e)**	$u = \sin x$ gives equal limits $0, 0$, so the integral and the average are 0.
23	**(a)**	Geometric $\frac{2}{e-1}$ plus telescoping $\frac{1}{2}$ gives $\frac{3+e}{2e-2}$.
24	**(b)**	Integrand $2 + x^{1/2} - x^{-2}$; value $\frac{292}{9} = 32\frac{4}{9}$.
25	**(c)**	$\frac{x}{9} \cdot \frac{1}{1+x^2/9}$ gives $\sum (-1)^n \frac{x^{2n+1}}{9^{n+1}}$.
26	**(b)**	$\sin^2 x = \frac{1-\cos 2x}{2}$, expand $\cos 2x$: $\frac{2}{2!}x^2 - \frac{2^3}{4!}x^4 + \cdots$.
27	**(d)**	Limit equals $F'(2)$ with $F'(x) = \sqrt{1+x^3}$, so the value is 3.
28	**(b)**	Ratio $= \frac{x^2}{4(n+1)(n+2)} \to 0$ for every x, so it converges for all x.